Year 13 Further Maths - Regression

Similar questions - Least squares lines, coding and reliability

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A set of bivariate data (x, y) is summarised as follows.

$$n = 20,\ \sum x = 14.010,\ \sum y = 77.520,\ \sum x^2 = 11.766,\ \sum y^2 = 315.918,\ \sum xy = 59.431.$$

(a) Calculate the product-moment correlation coefficient. **[1]**

(b) Calculate the equation of the regression line of y on x, giving a and b to 3 s.f. **[3]**

(c) It is desired to estimate y when $x = 0.75$. State one reason for believing that the estimate will be reliable. **[1]**

(d) State what further information is needed to determine whether the estimate is reliable. **[1]**

2 At a resort, the number of ice-creams sold x (in hundreds) and the temperature y°F were recorded on 20 random days.

$$\sum x = 176.6,\ \sum x^2 = 1758.08,\ \sum y = 1408,\ \sum y^2 = 99992,\ \sum xy = 12838.6.$$

(a) Calculate the equation of the least squares regression line of y on x, in the form $y = a + bx$. **[3]**

(b) Explain the significance for the regression line of the quantity $\sum [y_i - (a + bx_i)]^2$. **[1]**

(c) It is decided to measure the temperature in degrees Celsius c instead, where $c = \frac{5}{9}(f - 32)$. Find the equation of the new regression line of c on x. **[2]**

3 The director of a concert hall wishes to investigate whether the price £P of the most expensive tickets affects the audience size H (in hundreds). Data are collected for 15 concerts (three at each price).

P (£)	75	65	55	45	35
H (hundred)	27, 27, 26	27, 25, 24	22, 21, 20	19, 18, 16	14, 12, 10

$$n = 15,\ \sum p = 825,\ \sum h = 308,\ \sum p^2 = 48375,\ \sum h^2 = 6770,\ \sum ph = 18050.$$

(a) Calculate the equation of the regression line of h on p. **[2]**

(b) State what change, if any, there would be to your answer if H had been measured in thousands. **[1]**

(c) For a special concert the most expensive tickets cost £50. Use your line to estimate the audience size, to the nearest hundred. **[1]**

(d) Comment on the reliability of your estimate, referring to the correlation coefficient for the data, which is 0.960, the value £50, and one other relevant factor. **[4]**

4 For a set of 12 points (x_i, y_i) and constants a and b it can be shown that

$$\sum (y_i - a - bx_i)^2 = 12(9 - a - 4b)^2 + 120(b - 0.4)^2 + 9.6.$$

(a) Explain why $\sum (y_i - a - bx_i)^2$ is minimised by taking $b = 0.4$ and $a = 7.4$, and write down the regression line of y on x. **[3]**

(b) State which of the words *dependent, independent, controlled, response* cannot apply to x if the line is to be used to estimate y. **[1]**

(c) Use the line to estimate y when $x = 10$ and state what must be true of $x = 10$ for the estimate to be reliable. **[2]**

(d) Variables u and v are related to x and y by $u = 3 + 5x$ and $v = 4 - 2y$. Show that the regression line of v on u has gradient -0.16. **[3]**

Question 1

(a) $S_{xy} = 59.431 - \dfrac{14.010 \times 77.520}{20} = 5.128$, $S_{xx} = 1.952$, $S_{yy} = 15.450$; $r = \dfrac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = 0.934$

(b) $b = \dfrac{S_{xy}}{S_{xx}} = 2.627$; $a = \bar{y} - b\bar{x} = 2.036$. $y = 2.036 + 2.627x$

(c) r is close to 1 (strong linear correlation), and $x = 0.75$ lies within the range of the data (interpolation)

(d) Whether x is independent (or controlled) and the y variable is the response; or the number of data points / scatter diagram to check that the relationship is linear

Question 2

(a) $b = \dfrac{S_{xy}}{S_{xx}} = 2.043$, $a = \bar{y} - b\bar{x} = 52.36$; $y = 52.36 + 2.043x$

(b) The regression line minimises this sum of squares of the vertical distances (residuals) between the data and the line

(c) $c = \frac{5}{9}(y - 32)$: $c = \frac{5}{9}(52.36 - 32) + \frac{5}{9}(2.043)x = 11.31 + 1.135x$

Question 3

(a) $h = 0.18 + 0.3700p$

(b) The gradient and intercept would both be divided by 10

(c) At $p = 50$: $h = 18.7$ so about 19 hundred, i.e. 1868 people

(d) $r = 0.960$ shows strong positive linear correlation (larger audiences at higher prices, possibly because the higher price concerts are more popular), which supports use of the line. 50 lies within the range of the prices (35 to 75), so this is interpolation and is reliable. Other factors: only five distinct prices; capacity limit (the hall can only hold about 2700); charity concerts may attract more people regardless of price

Question 4

(a) The squares $12(\cdot)^2$ and $120(\cdot)^2$ are non-negative and are zero when $9 - a - 4b = 0$ and $b = 0.4$; so the sum is minimised there ($b = 0.4$, $a = 9 - 1.6 = 7.4$). Line: $y = 7.4 + 0.4x$

(b) x is used to estimate y, so x cannot be the *dependent* or *response* variable

(c) $y = 7.4 + 4 = 11.4$; $x = 10$ should lie within the range of observed x values

(d) $x = \dfrac{u-3}{5}$ and $y = \dfrac{4-v}{2}$. Then $\dfrac{4-v}{2} = 7.4 + 0.4 \cdot \dfrac{u-3}{5}$, so $v = -10.32 - 0.16u$: gradient -0.16